

ABC Technologies – CI/CD DevOps Project Solution Document

1. Project Overview

ABC Technologies aimed to modernize its application development and deployment process after acquiring a large offline retail business. The existing system suffered from low availability, poor scalability, performance issues, and slow deployment cycles. This project implements a DevOps-based CI/CD pipeline to automate build, test, packaging, and deployment processes.

2. Tools & Technologies Used

- 1 AWS EC2 – Infrastructure hosting
- 2 GitHub – Source code management
- 3 Jenkins – Continuous Integration
- 4 Maven – Build and dependency management
- 5 Docker – Application containerization
- 6 Ansible – Automated deployment and configuration management
- 7 Linux (Ubuntu) – Operating system

3. CI/CD Architecture Flow

Developer pushes code to GitHub. Jenkins pulls the source code, compiles it using Maven, runs unit tests, and packages the application into a WAR file. Jenkins then builds a Docker image using Tomcat and the WAR file. Ansible automates the deployment by managing Docker containers on the EC2 instance.

4. Jenkins Continuous Integration

Jenkins was configured with multiple jobs to compile, test, and package the Java Maven application. Each stage ensures code quality and build reliability before moving to deployment.

5. Docker Containerization

Docker was used to package the application along with Tomcat into a portable container. This ensures consistency across environments and faster deployments.

6. Role of Ansible

Ansible acts as the deployment automation tool. It stops and removes any existing containers, builds a fresh Docker image from the Jenkins workspace, and deploys the application as a running container. This eliminates manual intervention and ensures repeatable deployments.

7. Business Benefits Achieved

- 1 High availability and scalability
- 2 Faster development and deployment cycles
- 3 Reduced human error through automation
- 4 Improved application performance
- 5 Lower time to market

8. Conclusion

The CI/CD pipeline successfully modernizes ABC Technologies' application delivery process. By integrating Jenkins, Docker, and Ansible, the organization achieves faster, reliable, and scalable deployments. Future enhancements may include Kubernetes orchestration and monitoring with Prometheus and Grafana.